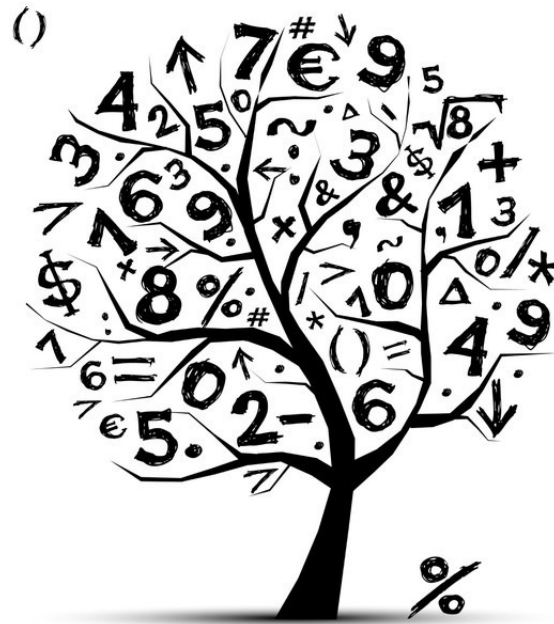


Last update: October 7, 2026

Absolute value



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1 Definition

Definition 1: absolute value

The **absolute value** $|a|$ of a real number a is the following non-negative real number:

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases} \quad (1)$$

From the definition of absolute value it immediately follows that:

$$|a| < \varepsilon \iff -\varepsilon < a < \varepsilon, \quad \forall \varepsilon > 0, a \in \mathbb{R} \quad (2)$$

and, similarly, for non-strict inequalities:

$$|a| \leq \varepsilon \iff -\varepsilon \leq a \leq \varepsilon, \quad \forall \varepsilon \geq 0, a \in \mathbb{R} \quad (3)$$

2 Triangle inequality

Observation 1: triangle inequality

$$|b + c| \leq |b| + |c|, \quad \forall b, c \in \mathbb{R} \quad (4)$$

Proof. We write the two relations:

$$-|b| \leq b \leq |b|, \quad -|c| \leq c \leq |c|$$

and add member to member:

$$-(|b| + |c|) \leq b + c \leq |b| + |c|$$

So for (3), with $\varepsilon = |b| + |c| \geq 0$ and $a = b + c$, it follows (4). $\square$

- The triangle inequality is also used in the following form:

$$|d - e| \leq |d - f| + |e - f| \quad \forall d, e, f \in \mathbb{R} \quad (5)$$

To obtain it, it suffices to set in (4):

$$b = d - f, \quad c = f - e$$

and we get:

$$|d - f + f - e| = |d - e| \leq |d - f| + |f - e| = |d - f| + |e - f|$$

since:

$$|f - e| = |e - f| \quad \forall e, f \in \mathbb{R}$$

- Finally, another form of the triangle inequality is:

$$|g| \leq |g - h| + |h| \quad \text{that is} \quad |g| - |h| \leq |g - h| \quad \forall g, h \in \mathbb{R} \quad (6)$$

To obtain it, it suffices to set in (4):

$$b = g - h, \quad c = h$$

Observation 2: reverse triangle inequality

$$||g| - |h|| \leq |g - h|, \quad \forall g, h \in \mathbb{R} \quad (7)$$

Proof. From (6), we have:

$$|g| - |h| \leq |g - h| \quad \forall g, h \in \mathbb{R}$$

Similarly exchanging g for h in (6) we get:

$$|h| - |g| \leq |h - g| = |g - h| \quad \text{that is} \quad |g| - |h| \geq -|g - h|$$

then we have

$$-(|g - h|) \leq |g| - |h| \leq |g - h| \quad \forall g, h \in \mathbb{R}$$

Accordingly from (3), with $\varepsilon = |g - h| \geq 0$ and $a = |g| - |h|$, we get the reverse triangle inequality. □

- The triangle inequality (4) can easily be generalized to the case of k addends:

$$\left| \sum_{i=1}^k b_i \right| \leq \sum_{i=1}^k |b_i|. \quad (8)$$

- The following immediate properties also hold:

$$|bc| = |b||c|, \quad \left| \frac{b}{c} \right| = \frac{|b|}{|c|}, \quad |-b| = |b| \quad \forall b, c \in \mathbb{R} \quad (c \neq 0 \text{ in the quotient}). \quad (9)$$